

BASS-II SCIENCE REQUIREMENTS

- General Requirements
 - 20 samples for each of the 5 PIs will be flown (including 20 clear pmma rods, 20 black pmma rods (varied diameters for both); 19 SIBAL samples, 1 4 cm pmma spherical section, 20 pmma films (varied thicknesses) and 20 pmma slabs (varied thicknesses and 1 sided or 2 sided holders). Also 3 Nomex-III samples.
 - 27 additional igniter wires will be flown. Solid fuel ignition uses a Kanthal hot wire.
 - ISS nitrogen gas will be used (0 to 500 SCCM N_2). Instead of connecting the Experiment Assembly to a gas bottle it is connected via the GN_2 hose to the MSG GN_2 interface. GN_2 is used to vitiate the MSG working volume to desired O_2 levels.
 - 8 additional flash cards will be flown for the still camera.
- Parameter Measurement
 - Measurement of the GN_2 flow rate, fan voltage, air flow velocity and direction, and flame radiation will be measured and overlaid on the Video.
 - The acquisition rate of 4 Hz can not be changed from SPICE/BASS.
 - The measurement accuracy of $\pm 10\%$ was met by the SPICE hardware prior to its launch.
- Imaging Requirements
 - The field of view of the video can not be changed from BASS.
 - The D300s still image requirements are the same as for BASS.

BASS-II SCIENCE REQUIREMENTS

- Acceleration
 - BASS-II testing will be scheduled around planned thruster firing or spacecraft dock or undock.
 - The acceleration environment will be measured by a nearby SAMS during BASS-II testing.
- MSG Working Volume Gas Concentrations
 - Project will include steps in crew procedures to install a Compound Specific Analyzer (CSA-CP) and the CO₂ sensor (CDM) into the MSG to measure the Oxygen concentration. These devices will provide readings before and after each burn of the MSG working volume well mixed gas concentrations. Calibration checks will utilize the Columbus and Node 3 O₂ and CO₂ readings.
- Crew support
 - Crew time is needed to conduct the BASS-II experiments. As part of the nominal ops, the crew establishes the desired test ambient atmosphere and flow. The crew manually inserts the fuel sample, adjusts the fuel location and igniter position within the camera fields of view. The crew manually ignites the sample and triggers the still camera. The crew performs flow calibration and soot cleaning as needed.
 - The crew works closely with the BASS-II PI team to determine test settings and provide real-time feedback during each burn.

BASS-II SUCCESS CRITERIA

- Minimal Success: 10 samples from each PI (of their total 20) will be successfully tested per PI matrices at the desired test conditions (O_2 , flow). Ignition is successful for these tests, and flame video data is obtained. Still camera data is obtained for at least 7 tests per PI. Valuable information informing SoFIE science requirements is obtained.
- Complete Success: All PI samples (100) are tested at the desired test conditions. Video data is obtained for all tests. Ignition occurs in most of the tests (90+). Extinction is observed for tests that seek to find a flammability limit. Most of the tests (90+) provide still camera data. Valuable information informing SoFIE science requirements is obtained.

BASS-II Hardware Manifest

The manifest includes the following items utilizing designs that have previously flown:

27 ignitors with 9 of each of the 3 different types assembled per S1183MFA0015, S1183MFA0020, and S1183MFA0025.

8 Flash Cards - 4 GB Capacity (Different vendor from original, same functionality)

15 SIBAL Samples (2 cm wide) – Previously Flown Design - S1183MFA1020

4 SIBAL Samples (1 cm wide) – Previously Flown Design - S1183MFA1025

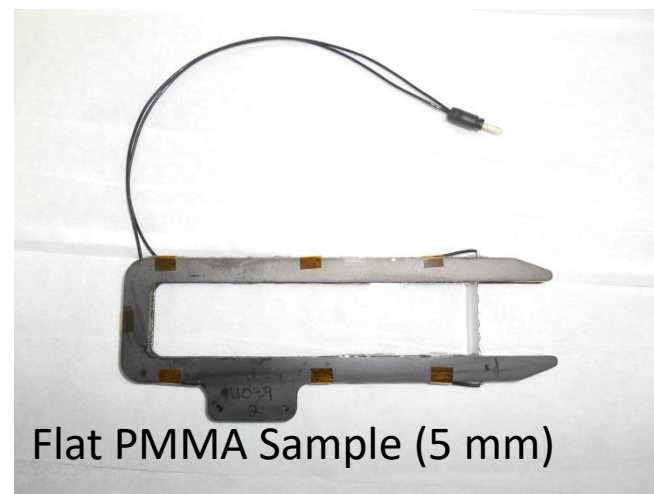
3 NOMEX Samples (2 cm wide) – Previously Flown Design - S1183MFA1055

BASS-II Hardware Manifest

The manifest includes the following newly designs hardware:

- 8 Flat 2 cm PMMA Samples (0.1 mm thick) - S1183MFA1110-1
- 4 Flat 2 cm PMMA Samples (0.2 mm thick) - S1183MFA1110-2
- 2 Flat 2 cm PMMA Samples (0.3, 0.4, 1.0, 2.0, 3.0, 4.0, and 5.0 mm thick) -
S1183MFA1110-3 thru S1183MFA1110-9
- 4 Flat 1 cm PMMA Samples (0.2 mm thick) - S1183MFA1115-1
- 1 Flat 1 cm PMMA Samples (1.0, 2.0, 3.0, 4.0, and 5.0 mm thick) –
S1183MFA1115-2 thru S1183MFA1110-6
- 14 Black PMMA Rod Samples (0.25" Diameter) - S1183MFA1090-1
- 6 Black PMMA Rod Samples (0.375" Diameter) - S1183MFA1090-2
- 2 Black PMMA Rod Samples (0.500" Diameter) - S1183MFA1090-3
- 12 Clear PMMA Rod Samples (0.25" Diameter) - S1183MFA1095-1
- 4 Clear PMMA Rod Samples (0.375" Diameter) - S1183MFA1095-2
- 2 Clear PMMA Rod Samples (0.500" Diameter) - S1183MFA1095-3
- 1 Flat 2 cm PMMA one-sided Samples (1.0, 2.0, 3.0, 4.0, and 5.0 mm thick) –
S1183MFA1120-1 thru S1183MFA1120-5
- 1 Sphere Sample 4 cm - S1183MFA1085

BASS-II SAMPLE PHOTOS



BASS-II Science Overview

Dr. Sandra Olson

BASS-II Science Overview

PS: Dr. Paul Ferkul, NCSE

PI: Dr. Sandra Olson, NASA GRC

Co-PI: Prof. Subrata Bhattacharjee, SDSU

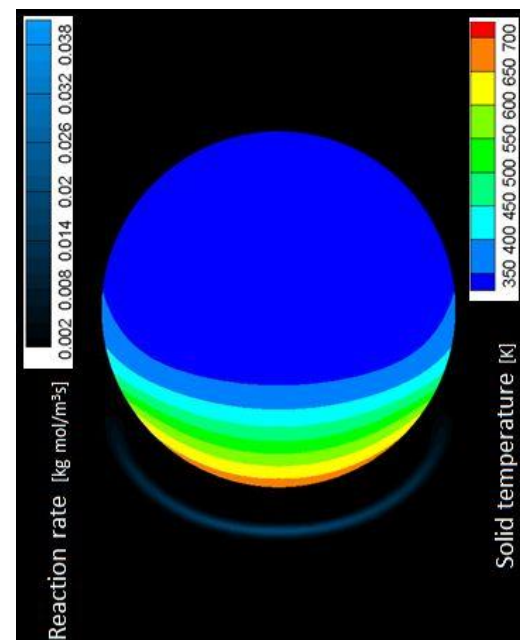
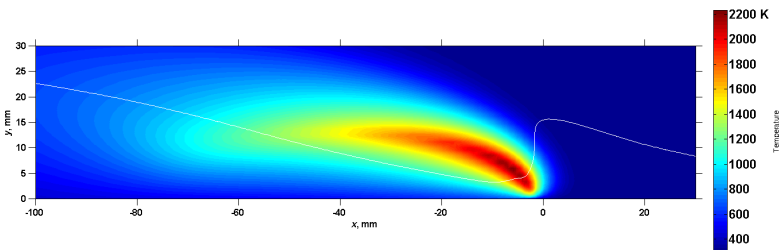
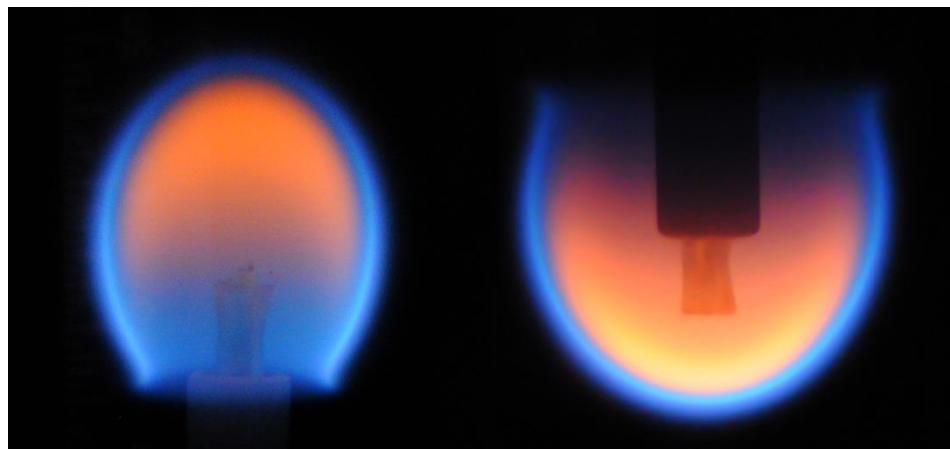
Co-PI: Prof. James T'ien, CWRU

Co-PI: Prof. Fletcher Miller, SDSU

Co-PI: Prof. Carlos Fernandez-Pello, UCB

PM: Robert Hawersaat, NASA GRC

Engineering Team: ZIN Technologies, Inc.



BASS-II Science Overview

Relevance/Impact: Burning and Suppression of Solids - II (BASS-II) tests the hypothesis that materials in microgravity, with adequate ventilation, burn as well if not better than the same material in normal gravity with other conditions being identical (pressure, oxygen concentration, temperature, etc.).

Objectives:

- To identify what is the worst case for material flammability in spacecraft environments, and how does that compare to the terrestrial upward burning used to screen the materials for safe use aboard spacecraft.
- To determine how quickly a fire in space can grow and if the flames reach a finite size or continue to grow. This has implications for firefighting strategies in spacecraft.

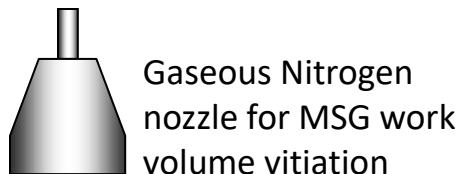
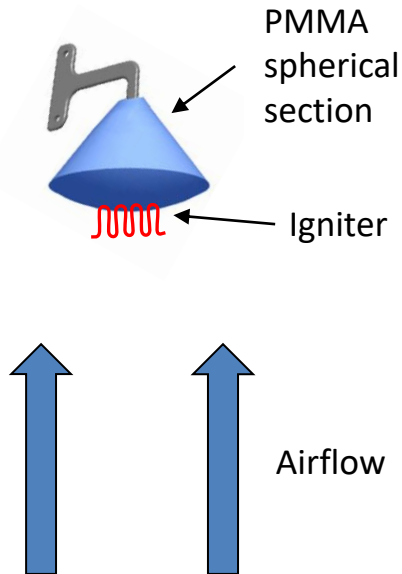
Primary variables: The main variables to be tested are the effects of ambient oxygen concentration, ventilation flow velocity, and fuel type, thickness, and geometry.

Development Approach:

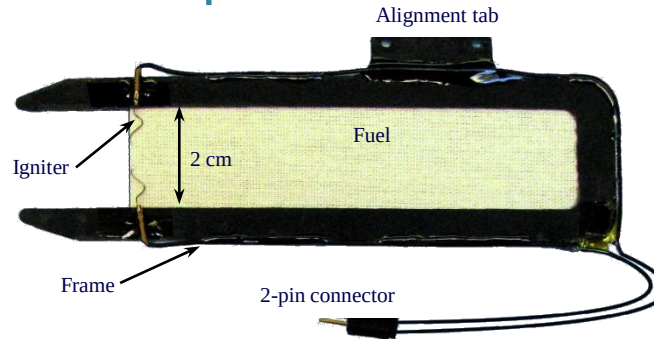
BASS-II will utilize the on orbit BASS hardware launched originally for SPICE on STS-126. New samples, igniters, and more flash cards will provide new data. Crew is required to set up and operate the experiment. Video and data down-linked to the ground for evaluation.

BASS-II Science Overview

BASS-II Assembly for PMMA Spherical section



BASS-II Fuel Geometries (not to scale)



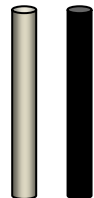
- 1 and 2-cm width fuel sheets
- PMMA
 - different thicknesses
 - 1 or 2 sided burns
 - SIBAL fabric
 - Nomex- III fabric



4-cm PMMA Spherical section (Forward stagnation region ignition and flame growth)



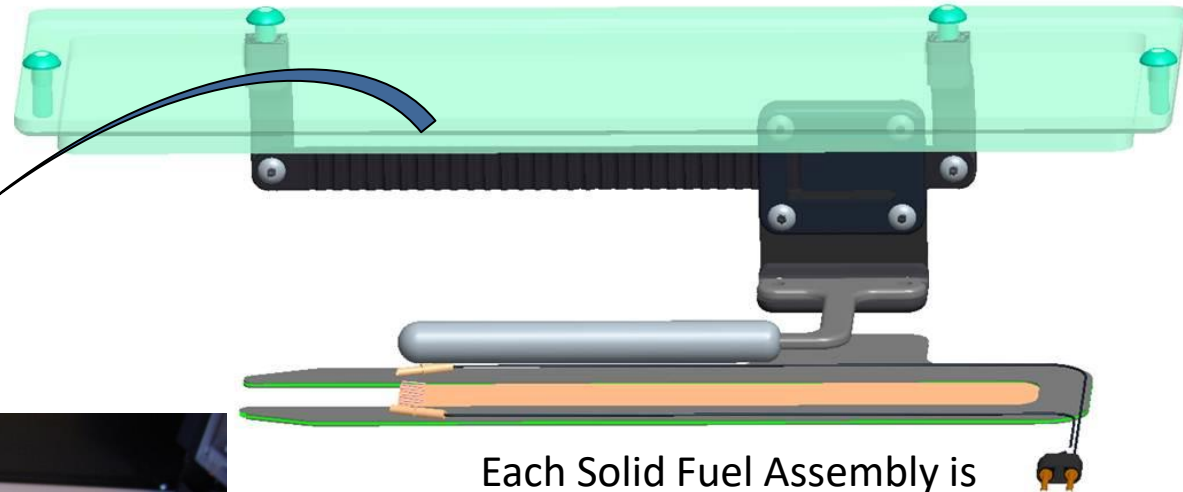
PMMA rods (black and clear), $\frac{1}{4}$ ", $\frac{3}{8}$ ", and $\frac{1}{2}$ " diameters



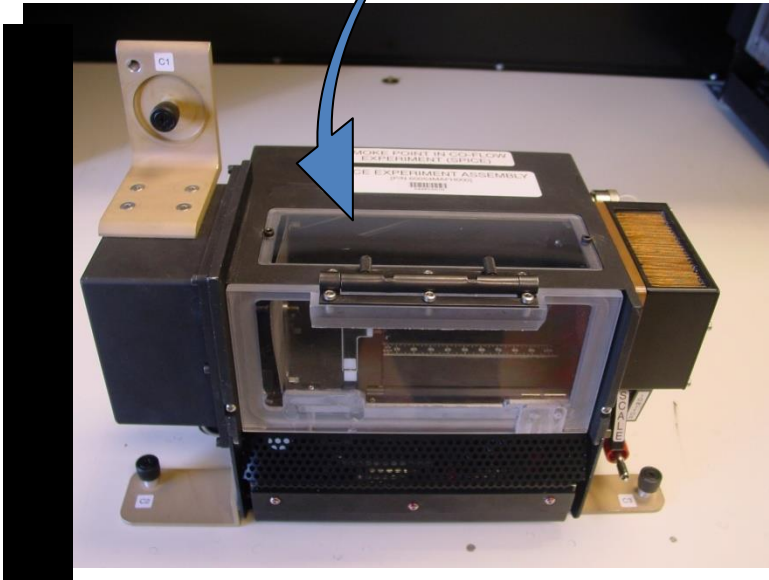
BASS-II Science Overview

Sample Mounting

The top window supports the solid fuel mounting rail.



Each Solid Fuel Assembly is mounted inside the Experiment Assembly via a magnetic interface such that it can be installed and removed by the crew



BASS-II Science Overview

priority order test matrix per PI: Olson

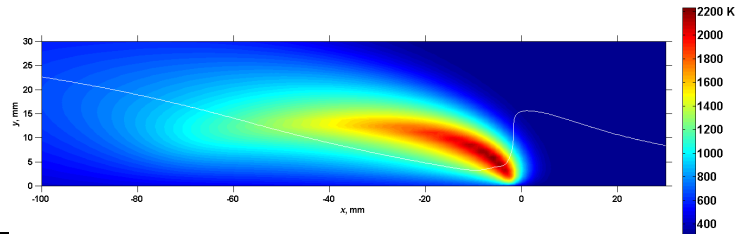


BASS- II test Matrix		PI:	Olson		
Priority Test #	O2	Flow(s)	Sample #/type/description	brief objective of test	notes
1	21	5, 3, 2, 1, min	1/4" rod, clear	steady spread at condition(s)	ignite at a consistent concurrent flow, then reduce to low flows and measure spread, repeat until extinguished. Determine solid response time.
2	21	5, etc	3/8" rod, clear	steady spread at condition(s)	diameter effect, timescale estimate
3	16	5, 3, 2, 1	1/4" rod, clear	find low velocity limit	wait at each flow for solid phase timescale
4	16	5, increase	1/4" rod, clear	find blowoff limit	can increase flow every 5-10 seconds
5	15	5, decrease	1/4" rod, clear	find low velocity limit	may re-use same rod for repeated tests, to preserve samples
6	15	5, increase	1/4" rod clear	find blowoff limit	can increase flow every 5-10 seconds
7	17 or 14	5, decrease	1/4" rod, clear	find low velocity limit	
8	17 or 14	5, increase	1/4" rod, clear	find blowoff limit	can increase flow every 5-10 seconds
9	21	5, etc	1/2" rod, clear	steady spread at condition(s)	diameter effect, timescale estimate
10	21	5,3,2,1, min	1/4" rod, black	steady spread at condition(s)	to determine if results depend on material color, for comparisons with Pello results
11	tbd	5, decrease	3/8" rod, clear	find low velocity limit	diameter effect
12	tbd	5, decrease	1/2" rod, clear	find low velocity limit	diameter effect
13	tbd	5, decrease	1/4" rod, black	find low velocity limit	color dependence on limit
14	tbd	tbd	1/4" rod, clear	tbd	possible repeat limit
15	tbd	5, decrease	3/8" rod, clear	find low velocity limit	diameter effect
16	tbd	5, increase	3/8" rod, clear	find blowoff limit	diameter effect
17	19	tbd	1/4" rod, clear	tbd	
18	19	tbd	1/4" rod, clear	tbd	
19	18	tbd	1/4" rod, clear	tbd	
20	18	tbd	1/4" rod, clear	tbd	

Notes in the event of a non-ignition of the rod at low O2, an opposed ignition will be attempted, followed by an ambient air opposed or concurrent test at TBD flow(s).

BASS-II Science Overview

priority order test matrix per PI: Bhattacharjee



BASS- II test Matrix PI: Bhattacharjee					
Priority					
Test #	O2	Flow(s)	Sample #/type/description	brief objective of test	notes
1	21	0 mm/s	A.1/PMMA/thickness 0.1 mm, width 20 mm	study thickness effect	
2	21	0 mm/s	B.1/PMMA/thickness 0.2 mm, width 20 mm	study thickness effect	
3	21	0 mm/s	D.1/PMMA/thickness 0.3 mm, width 20 mm	study thickness effect	
4	21	20 mm/s	A.2/PMMA/thickness 0.1 mm, width 20 mm	study boundary layer/flow effect	
5	21	0 mm/s	C.1/PMMA/thickness 0.2 mm, width 10 mm	study width effect	
6	15	0 mm/s	A.4/PMMA/thickness 0.1 mm, width 20 mm	study oxygen effect	
7	21	0 mm/s	E.1/PMMA/thickness 0.4 mm, width 20 mm	study thickness effect	
8	21	20 mm/s	B.2/PMMA/thickness 0.2 mm, width 20 mm	study boundary layer/thickness/flow effect	
9	21	50 mm/s	A.3/PMMA/thickness 0.1 mm, width 20 mm	study boundary layer/flow effect	
10	21	50 mm/s	C.3/PMMA/thickness 0.2 mm, width 10 mm	study width/flow effect	
11	21	50 mm/s	B.3/PMMA/thickness 0.2 mm, width 20 mm	study boundary layer/thickness effect	
12	21	50 mm/s	D.2/PMMA/thickness 0.3 mm, width 20 mm	study boundary layer/thickness effect	
13	21	50 mm/s	E.2/PMMA/thickness 0.4 mm, width 20 mm	study boundary layer/thickness effect	
14	21	50 mm/s	C.2/PMMA/thickness 0.2 mm, width 10 mm	study oxygen/width effect	
15	15	20 mm/s	A.5/PMMA/thickness 0.1 mm, width 20 mm	study oxygen/flow/thickness effect	
16	15	50 mm/s	A.6/PMMA/thickness 0.1 mm, width 20 mm	study oxygen/flow/thickness effect	
17	15	50 mm/s	B.4/PMMA/thickness 0.2 mm, width 20 mm	study oxygen/flow/thickness effect	
18	15	50 mm/s	C.4/PMMA/thickness 0.2 mm, width 10 mm	study oxygen/width effect	
19	21	0 mm/s	A.7/PMMA/thickness 0.1 mm, width 20 mm	test data repeatability	
20	21	20 mm/s	A.8/PMMA/thickness 0.1 mm, width 20 mm	test data repeatability	

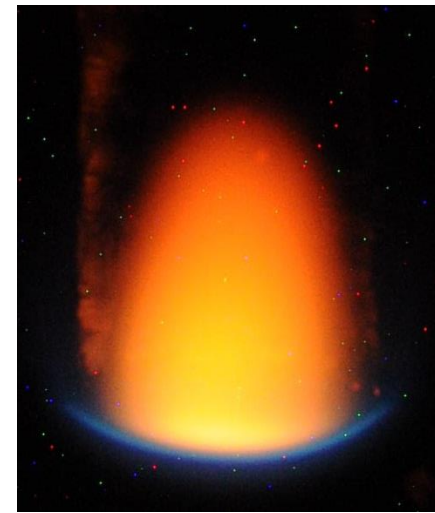
BASS-II Science Overview

priority order test matrix per PI: T'ien

BASS- II test Matrix		PI:	T'ien		
Priority Test #	O2	Flow(s)	Sample #/type/description	brief objective of test	notes
1	17	10	2 cm, concurrent SIBAL	steady spread at condition(s)	
2	17	10, decr	2 cm, concurrent SIBAL	low flow limit	
3	17	10	pmma sphere section	ignition test	also growth rate
4	17	10, incr	2 cm, concurrent SIBAL	blowoff limit	
5	18	10, decr	2 cm, concurrent SIBAL	low flow limit	
6	18	10, incr	2 cm, concurrent SIBAL	blowoff limit	
7	19	10, decr	2 cm, concurrent SIBAL	low flow limit	
8	19	10, incr	2 cm, concurrent SIBAL	blowoff limit	
9	16 or 20	10, decr	2 cm, concurrent SIBAL	low flow limit	
10	16 or 20	10, incr	2 cm, concurrent SIBAL	blowoff limit	
11	16 or 20	10	2 cm, concurrent SIBAL	steady spread at condition(s)	
12	16 or 20	tbd	1 cm SIBAL		
13	18	10	2 cm, concurrent SIBAL	steady spread at condition(s)	
14	18	tbd	1 cm SIBAL		
15	19	10	2 cm, concurrent SIBAL	steady spread at condition(s)	
16	19	tbd	1 cm SIBAL		
17	tbd	10, decr	2 cm, opposed SIBAL	low flow limit	
18	tbd	10, incr	2 cm, opposed SIBAL	blowoff limit	
19	tbd	tbd	2 cm, SIBAL		potential repeat of limit
20	TBD	tbd	1 cm SIBAL		

BASS-II Science Overview

priority order test matrix per PI: Miller



Priority	Sample #	width (cm)	thickness (mm)	backing	flow velocity (cm/s)	Direction	oxygen %	# sides to burn	Notes
1	15	2	5	mica	10 (ramp down)	opposed	21	one	need mica out the back
2	13	2	3	mica	10 (ramp down)	opposed	21	one	need mica out the back
3	11	2	1	mica	10 (ramp down)	opposed	21	one	need mica out the back
4	10	2	5	none	10 (ramp down)	opposed	21	two	
5	8	2	3	none	10 (ramp down)	opposed	21	two	
6	6	2	1	none	10 (ramp down)	opposed	21	two	
	1	1	1	none	10 (ramp down)	opposed	21	two	
	2	1	2	none	10 (ramp down)	opposed	21	two	
	3	1	3	none	10 (ramp down)	opposed	21	two	
	4	1	4	none	10 (ramp down)	opposed	21	two	
	5	1	5	none	10 (ramp down)	opposed	21	two	
	7	2	2	none	10 (ramp down)	opposed	21	two	
	9	2	4	none	10 (ramp down)	opposed	21	two	
	12	2	2	mica	10 (ramp down)	opposed	21	one	need mica out the back
	14	2	4	mica	10 (ramp down)	opposed	21	one	need mica out the back
	16	2	1	none	10 (ramp down)	opposed	21	two	extra; use for thicker samples if possible
	17	2	2	none	10 (ramp down)	opposed	21	two	extra; use for thicker samples if possible
	18	2	3	none	10 (ramp down)	opposed	21	two	extra; use for thicker samples if possible
	19	2	4	none	10 (ramp down)	opposed	21	two	extra; use for thicker samples if possible
	20	2	5	none	10 (ramp down)	opposed	21	two	extra; use for thicker samples if possible

BASS-II Science Overview

priority order test matrix per PI: Fernandez-Pello



BASS- II test Matrix PI: Fernandez-Pello					
Priority Test #	O2	Flow(s)	Sample #/type/description	brief objective of test	notes
1	21	5, 3, 2, 1, min	1/4" rod, black	steady spread at condition(s)	ignite at a consistent OPPOSED flow, then reduce to low flows and measure spread, repeat until extinguished. Determine solid response time.
2	21	5, etc	3/8" rod, black	steady spread at condition(s)	diameter effect, timescale estimate
3	16	5, 3, 2, 1	1/4" rod, black	find low velocity limit	wait at each flow for solid phase timescale
4	16	5, increase	1/4" rod, black	find blowoff limit	can increase flow every 5-10 seconds
5	15	5, decrease	1/4" rod, black	find low velocity limit	may re-use same rod for repeated tests, to preserve samples
6	15	5, increase	1/4" rod black	find blowoff limit	can increase flow every 5-10 seconds
7 17 or 14	5, decrease	1/4" rod, black	find low velocity limit		
8 17 or 14	5, increase	1/4" rod, black	find blowoff limit	can increase flow every 5-10 seconds	
9	21	5, etc	1/2" rod, black	steady spread at condition(s)	diameter effect, timescale estimate
10	21	5,3,2,1, min	1/4" rod, black	steady spread at condition(s)	
11 tbd	5, decrease	3/8" rod, black	find low velocity limit	diameter effect	
12 tbd	5, decrease	1/2" rod, black	find low velocity limit	diameter effect	
13 tbd	5, decrease	1/4" rod, black	find low velocity limit		
14 tbd	tbd	1/4" rod, black	tbd	possible repeat limit	
15 tbd	5, decrease	3/8" rod, black	find low velocity limit	diameter effect	
16 tbd	5, increase	3/8" rod, black	find blowoff limit	diameter effect	
17	19 tbd	1/4" rod, black	tbd		
18	19 tbd	1/4" rod, black	tbd		
19	18 tbd	1/4" rod, black	tbd		
20	18 tbd	1/4" rod, black	tbd		